AER320 Term Test 1: practice set

Covers HS ch. 1, 2 and 4.1 to 4.9.

How to use this. Part A is a 10-minute concept check. Part B is eleven problems in roughly the order the chapters run. For a timed mock test, do **Problems 1, 4, 6, 8 and 10(a) in 55 minutes** with only your formula sheet, then mark against the answer key on page 14 before reading the worked solutions. Every problem has a figure, and the statement also gives the geometry in coordinates or angles. Try sketching your own diagram from the statement before looking at the figure; on the real test that sketch is half the marks.

Each problem starts on its own page, with the rest of the page left for your working.

Units SI, $g = 9.81$ m/s^2, final answers to 4 significant figures.

Part A: concept check (10 min)

A1. A force can slide along its line of action without changing its external effect on a rigid body. The force is therefore a
(a) free vector (b) sliding vector (c) fixed vector (d) unit vector

A2. The moment of a couple is a (a) free vector (b) sliding vector (c) fixed vector. Why does that make couples easy to deal with in a force-couple reduction?

A3. $\vec{r} \times \vec{F} = \vec{0}$, where $\vec{r}$ runs from O to a point on the line of action of a non-zero $\vec{F}$. What can you conclude?

A4. A force has $\alpha = 60^\circ$ and $\beta = 45^\circ$. What are the possible values of γ?

A5. A system of parallel forces can always be reduced to
(a) a single force (b) a single couple (c) a single force, or a single couple if the forces sum to zero
(d) a wrench with non-zero pitch

A6. $\vec{F} = \{20\hat{\jmath}\}$ N acts at the point $(2, 0, 4)$ m. Without a determinant, state the moment about the y axis. Then find the moments about the x and z axes.

A7. A load rises linearly from 0 to 6 kN/m over 3 m. State the resultant and where it acts.

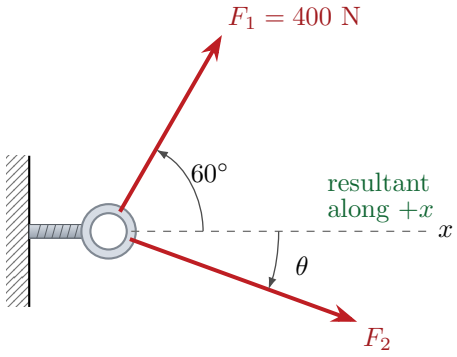
A8. The same force system is reduced to a force-couple system at A, then separately at B. Which of $\vec{F}_R$ and $\vec{M}_R$ comes out the same both times?

A9. $\vec{F} \cdot \hat{u}_a$ comes out negative. What does that tell you?

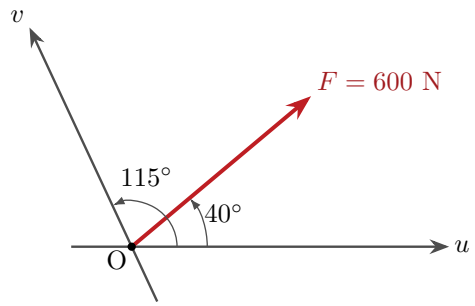
A10. In $M_a = \hat{u}_a \cdot (\vec{r} \times \vec{F})$, where can $\vec{r}$ start and end?

Part B: problems

Problem 1 *Parallelogram law, law of sines (HS 2.3)*
Two forces act on a hook. $F_1 = 400$ N is directed $60°$ above the $+x$ axis. F_2 is directed at an angle θ below the $+x$ axis. (a) If $\theta = 20°$, find F_2 so that the resultant acts along the $+x$ axis, and find the resultant's magnitude. (b) What θ makes F_2 as small as possible while the resultant still acts along $+x$? Give that F_2 and the resultant.

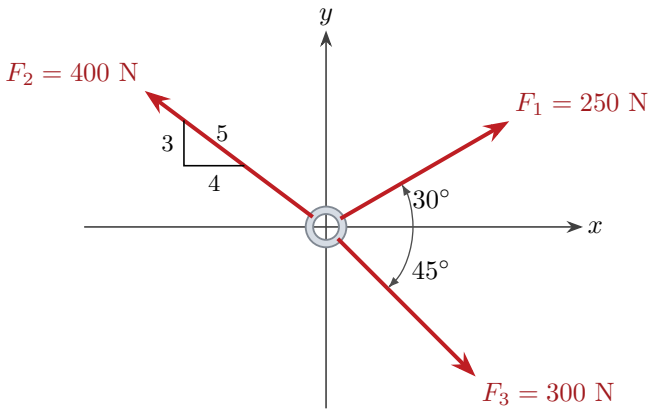


Problem 2 *Components along non-perpendicular axes (HS 2.3, 2.9)*
$F = 600$ N acts at 40° counter-clockwise from the u axis. The v axis is at 115° counter-clockwise from the u axis. (a) Resolve $\vec{F}$ into components acting along u and v. (b) Find the projection of $\vec{F}$ onto the v axis and explain why it is not the same as the v component from (a).



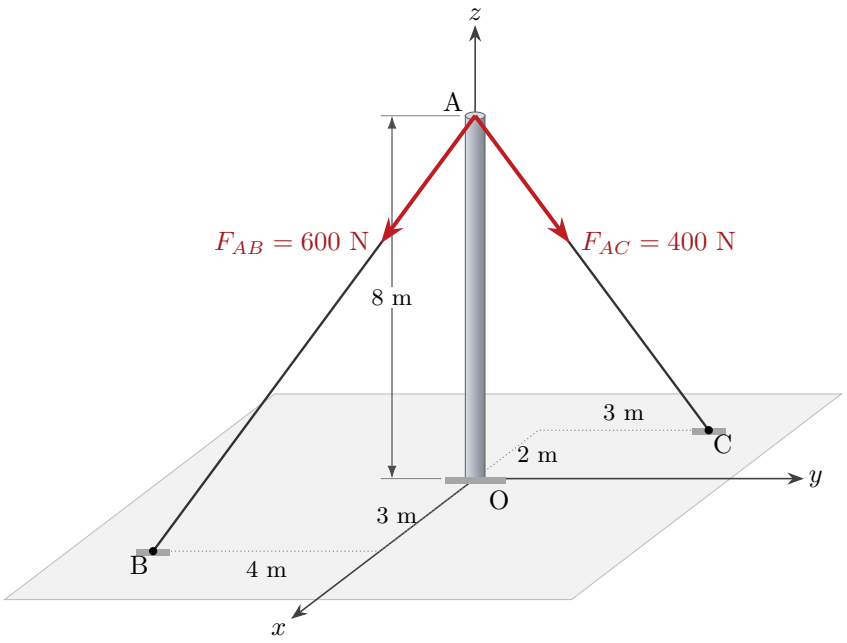
Problem 3 *Coplanar resultant (HS 2.4)*

Three forces act at a point: $F_1 = 250$ N at 30° above $+x$; $F_2 = 400$ N up and to the left along a 3-4-5 slope (4 horizontal, 3 vertical); $F_3 = 300$ N at 45° below $+x$. Find the magnitude of the resultant and its direction measured counter-clockwise from $+x$.



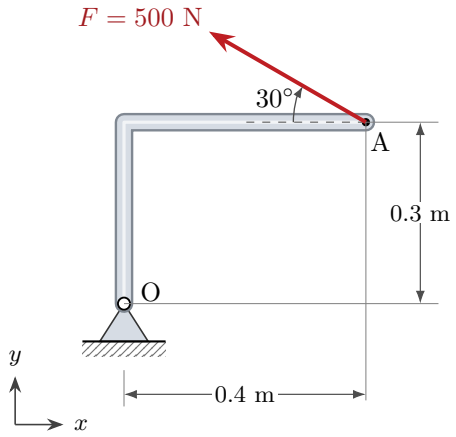
Problem 4 *Force along a line, dot product (HS 2.5 to 2.9)*

A mast has its top at A$(0, 0, 8)$ m. Cable AB runs to anchor B$(3, -4, 0)$ m with tension 600 N, and cable AC to anchor C$(-2, 3, 0)$ m with tension 400 N. (a) Write $\vec{F}_{AB}$ (the force on the mast) as a Cartesian vector and find its coordinate direction angles. (b) Find the resultant of the two cable forces at A. (c) Find the angle between the two cables. (d) Find the components of $\vec{F}_{AB}$ parallel and perpendicular to cable AC.



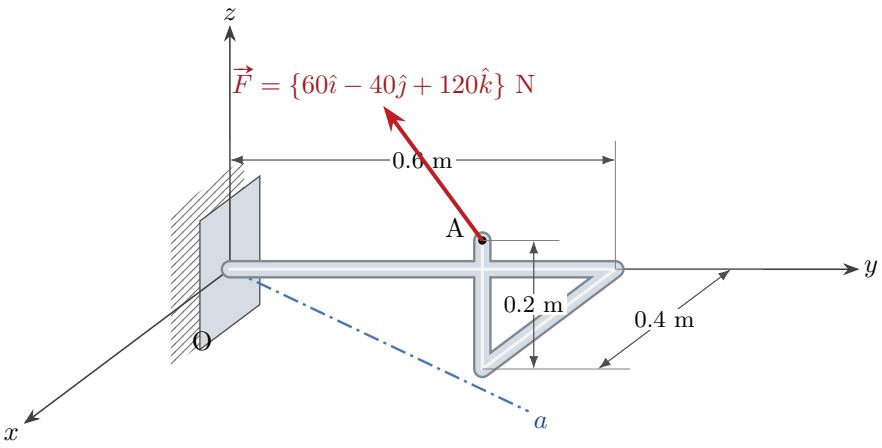
Problem 5 *Moment in 2D (HS 4.1 to 4.4)*

A 500 N force acts at point A, which sits at $(0.4, 0.3)$ m relative to O. The force points up and to the left at 30° above the $-x$ direction. (a) Find M_O using Varignon's theorem. (b) Find the perpendicular distance from O to the line of action. (c) Find the smallest force that, applied at A, gives a moment of 400 N·m *clockwise* about O. Give its magnitude and direction.



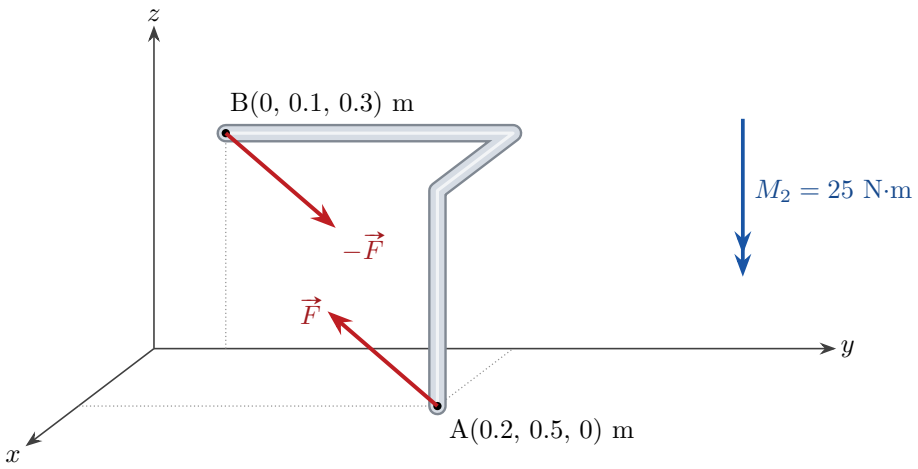
Problem 6 *Moment in 3D and about an axis (HS 4.3, 4.5)*

$\vec{F} = \{60\hat{\imath} - 40\hat{\jmath} + 120\hat{k}\}$ N acts at A$(0.4, 0.6, 0.2)$ m. (a) Find $\vec{M}_O$ and its magnitude. (b) Find the moment of $\vec{F}$ about the axis Oa, which lies in the xy plane and passes through O and the point $(3, 4, 0)$ m. Give it as a scalar and as a vector. (c) What is the moment about the y axis?



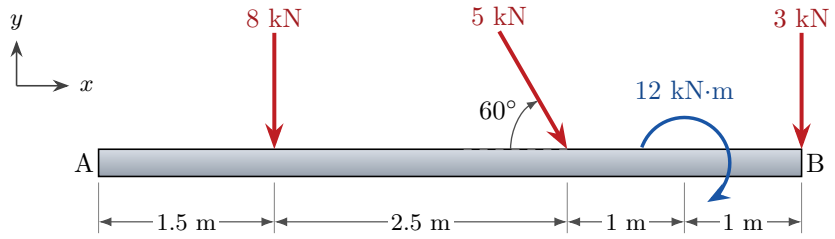
Problem 7 *Couples in 3D (HS 4.6)*

$\vec{F} = \{40\hat{\imath} - 30\hat{\jmath} + 60\hat{k}\}$ N acts at A$(0.2, 0.5, 0)$ m and $-\vec{F}$ acts at B$(0, 0.1, 0.3)$ m. A second couple of 25 N·m acts in the $-z$ direction. Find the resultant couple moment: vector, magnitude and coordinate direction angles.



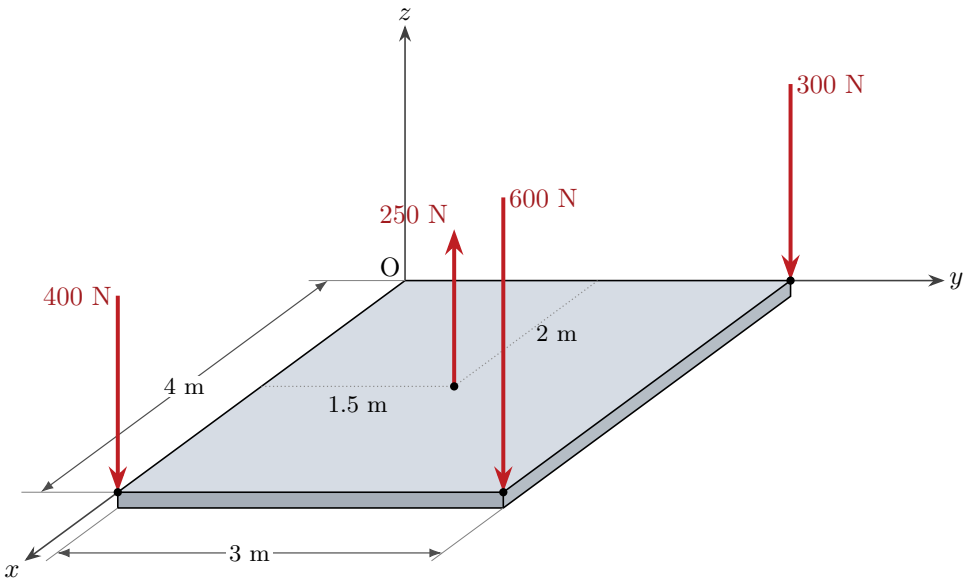
Problem 8 *Equivalent system and single resultant on a beam (HS 4.7, 4.8)*

A 6 m horizontal beam AB has A at the left end. Loads, all on the beam's axis: 8 kN down at 1.5 m from A; 5 kN at 4 m from A, directed down and to the right at 60° below the horizontal; a 12 kN·m clockwise couple at 5 m; 3 kN down at B. (a) Replace the loading with an equivalent force and couple at A. (b) Reduce it to a single resultant force and find where its line of action crosses the beam.



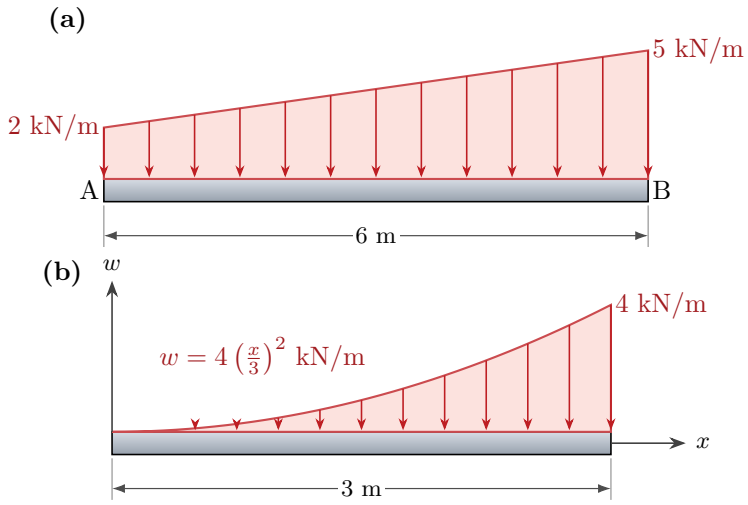
Problem 9 *Parallel forces on a slab (HS 4.8)*

A 4 m (x) by 3 m (y) slab lies in the xy plane with a corner at O. Vertical forces: 400 N down at $(4, 0)$, 600 N down at $(4, 3)$, 300 N down at $(0, 3)$ and 250 N *up* at $(2, 1.5)$ (m). Find the single resultant force and the point (x, y) where it acts.

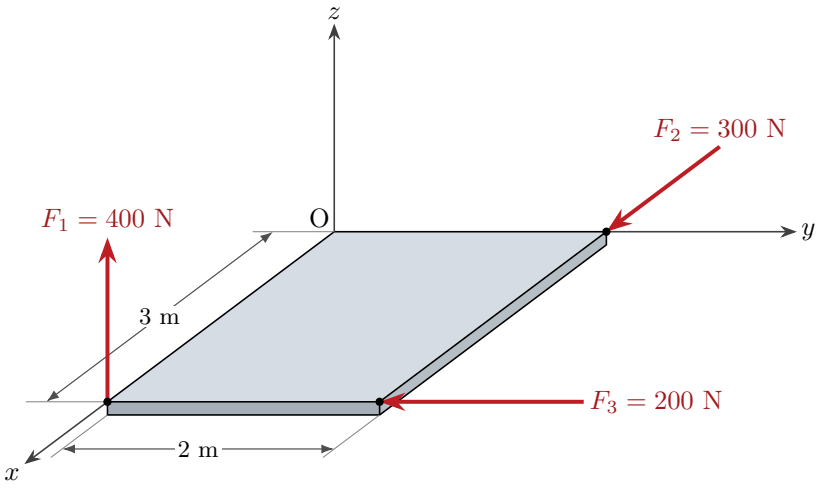


Problem 10 *Distributed loads (HS 4.9)*

(a) A 6 m beam carries a load rising linearly from 2 kN/m at A to 5 kN/m at B. Find the resultant and its distance from A. (b) A 3 m beam carries $w = 4(x/3)^2$ kN/m, x in m measured from the left end. Find the resultant and its location by integration.



Problem 11 *Wrench (HS 4.8, general case)*
$\vec{F}_1 = \{400\hat{k}\}$ N at $(3, 0, 0)$ m, $\vec{F}_2 = \{300\hat{\imath}\}$ N at $(0, 2, 0)$ m and $\vec{F}_3 = \{-200\hat{\jmath}\}$ N at $(3, 2, 0)$ m. Reduce the system to a wrench: find the resultant force, the wrench couple moment, and the point (x, y) where the wrench axis crosses the xy plane.



Answer key

Part A

A1. (b) sliding vector.
A2. (a) free vector: it can be added straight into $\sum \vec{M}$ at any reduction point, with no position vector.
A3. The line of action of $\vec{F}$ passes through O (so $\vec{r} \parallel \vec{F}$).
A4. $\cos^2\gamma = 1 - 0.25 - 0.5 = 0.25$, so $\gamma = 60°$ or $120°$.
A5. (c).
A6. $M_y = 0$, since the force is parallel to the y axis. $\vec{M}_O = (2,0,4) \times (0,20,0) = \{-80\hat{\imath} + 40\hat{k}\}$ N·m, so $M_x = -80$ N·m and $M_z = 40$ N·m.
A7. 9 kN, acting 2 m from the zero end (1 m from the 6 kN/m end).
A8. $\vec{F}_R$ is identical; $\vec{M}_R$ generally differs, by $\vec{r} \times \vec{F}_R$.
A9. The component of $\vec{F}$ along the line points opposite to $\hat{u}_a$ (angle over 90°).
A10. From any point on the axis to any point on the line of action of $\vec{F}$.

Part B

1	(a) $F_2 = 1013$ N, $F_R = 1152$ N (b) $\theta = 90°$, $F_2 = 346.4$ N, $F_R = 200.0$ N
2	(a) $F_u = 639.5$ N, $F_v = 425.5$ N (b) $\vec{F} \cdot \hat{u}_v = 155.3$ N
3	$F_R = 187.5$ N at 54.60° above $+x$
4	(a) $\{190.8\hat{\imath} - 254.4\hat{\jmath} - 508.8\hat{k}\}$ N; $\alpha = 71.46°$, $\beta = 115.1°$, $\gamma = 148.0°$ (b) $\{99.63\hat{\imath} - 117.6\hat{\jmath} - 873.5\hat{k}\}$ N, 887.0 N (c) 56.24° (d) 333.4 N, 498.8 N
5	(a) 229.9 N·m CCW (b) 0.4598 m (c) 800.0 N at 53.13° below $+x$
6	(a) $\{80.00\hat{\imath} - 36.00\hat{\jmath} - 52.00\hat{k}\}$ N·m, 102.0 N·m (b) 19.20 N·m, $\{11.52\hat{\imath} + 15.36\hat{\jmath}\}$ N·m (c) −36.00 N·m
7	$\{15.00\hat{\imath} - 24.00\hat{\jmath} - 47.00\hat{k}\}$ N·m, 54.86 N·m; 74.13°, 115.9°, 148.9°
8	(a) $\{2.500\hat{\imath} - 15.33\hat{\jmath}\}$ kN (15.53 kN, 80.74° below $+x$), $M_A = 59.32$ kN·m CW (b) crosses at 3.870 m from A
9	1050 N down at $(3.333, 2.214)$ m
10	(a) 21.00 kN at 3.429 m from A (b) 4.000 kN at 2.250 m
11	$\vec{F}_R = \{300\hat{\imath} - 200\hat{\jmath} + 400\hat{k}\}$ N (538.5 N); $M_{\parallel} = -445.7$ N·m (opposite $\vec{F}_R$); $(3.414, 0.6207)$ m

Worked solutions

Solution 1

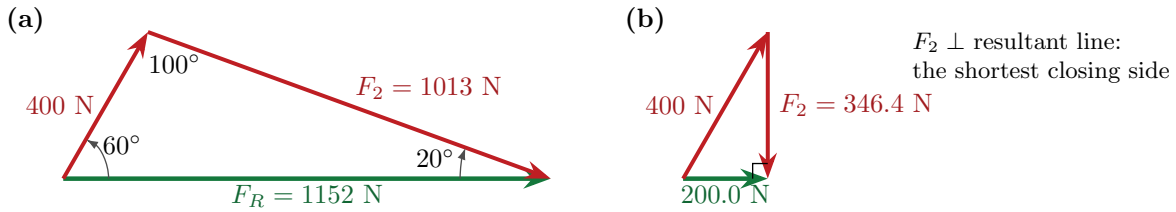
Draw the force triangle tip to tail with $\vec{F}_R$ along $+x$. The angle between $\vec{F}_1$ and $\vec{F}_R$ is 60° and between $\vec{F}_2$ and $\vec{F}_R$ is 20°, so the angle opposite $\vec{F}_R$ is $180^\circ - 60^\circ - 20^\circ = 100^\circ$.

(a) Law of sines:

$$\frac{F_2}{\sin 60^\circ} = \frac{400}{\sin 20^\circ} \;\Rightarrow\; F_2 = \frac{400(0.8660)}{0.3420} = 1013 \text{ N}, \qquad F_R = \frac{400 \sin 100^\circ}{\sin 20^\circ} = 1152 \text{ N}$$

Component check: $\sum F_y = 400 \sin 60^\circ - F_2 \sin 20^\circ = 346.4 - 346.4 = 0$. Good.

(b) $\vec{F}_2$ only has to cancel the 346.4 N vertical component of $\vec{F}_1$. The shortest vector that does this is vertical, so $\theta = 90^\circ$ and $F_2 = 346.4$ N. The resultant is then just $F_1 \cos 60^\circ = 200.0$ N.

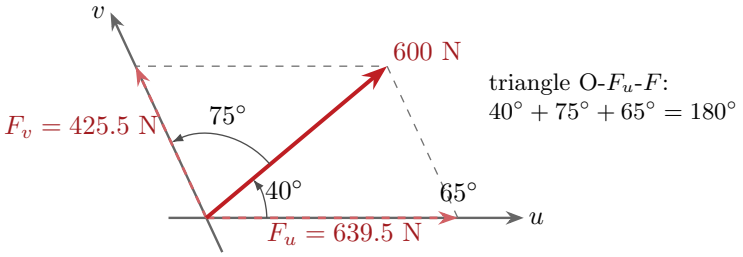


Solution 2

(a) The components along u and v form a triangle with $\vec{F}$. The angle between $\vec{F}$ and u is 40°, between $\vec{F}$ and v is $115^\circ - 40^\circ = 75^\circ$, and the angle opposite $\vec{F}$ is $180^\circ - 115^\circ = 65^\circ$.

$$F_u = \frac{600 \sin 75^\circ}{\sin 65^\circ} = 639.5 \text{ N}, \qquad F_v = \frac{600 \sin 40^\circ}{\sin 65^\circ} = 425.5 \text{ N}$$

(b) Projection $= F \cos(\text{angle between } \vec{F} \text{ and } v) = 600 \cos 75^\circ = 155.3$ N. The two differ because u and v are not perpendicular: a component comes from the parallelogram, a projection comes from dropping a perpendicular. They agree only for orthogonal axes. When a test asks for the *components along* u and v, it wants the parallelogram.



Solution 3

$$\vec{F}_1 = 250(\cos 30^\circ,\ \sin 30^\circ) = (216.5,\ 125.0) \text{ N}$$
$$\vec{F}_2 = 400(-\tfrac{4}{5},\ \tfrac{3}{5}) = (-320.0,\ 240.0) \text{ N}$$
$$\vec{F}_3 = 300(\cos 45^\circ,\ -\sin 45^\circ) = (212.1,\ -212.1) \text{ N}$$
$$F_{Rx} = 108.6 \text{ N}, \qquad F_{Ry} = 152.9 \text{ N}$$
$$F_R = \sqrt{108.6^2 + 152.9^2} = 187.5 \text{ N}, \qquad \theta = \tan^{-1}(152.9/108.6) = 54.60^\circ$$

Both components are positive, so the resultant is in the first quadrant.

Solution 4

(a) Tip minus tail: $\vec{r}_{AB} = (3, -4, -8)$ m, $|\vec{r}_{AB}| = \sqrt{89} = 9.434$ m.

$$\vec{F}_{AB} = 600\,\frac{(3,-4,-8)}{9.434} = \{190.8\hat{\imath} - 254.4\hat{\jmath} - 508.8\hat{k}\}\ \text{N}$$

$\alpha = \cos^{-1}(3/9.434) = 71.46°$, $\beta = \cos^{-1}(-4/9.434) = 115.1°$, $\gamma = \cos^{-1}(-8/9.434) = 148.0°$.

(b) $\vec{r}_{AC} = (-2, 3, -8)$ m, $|\vec{r}_{AC}| = \sqrt{77} = 8.775$ m, so $\vec{F}_{AC} = 400(-2, 3, -8)/8.775 = \{-91.17\hat{\imath} + 136.8\hat{\jmath} - 364.7\hat{k}\}$ N.

$$\vec{F}_R = \{99.63\hat{\imath} - 117.6\hat{\jmath} - 873.5\hat{k}\}\ \text{N}, \qquad F_R = 887.0\ \text{N}$$

(c) $\cos\theta = \hat{u}_{AB} \cdot \hat{u}_{AC} = \dfrac{(3)(-2) + (-4)(3) + (-8)(-8)}{9.434(8.775)} = \dfrac{46}{82.78} = 0.5557$, so $\theta = 56.24°$.

(d) $F_{\parallel} = \vec{F}_{AB} \cdot \hat{u}_{AC} = 600(0.5557) = 333.4$ N and $F_{\perp} = \sqrt{600^2 - 333.4^2} = 498.8$ N.

Solution 5

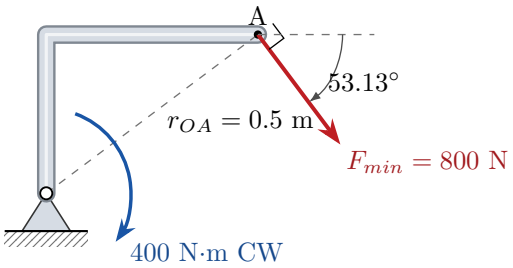
(a) $\vec{F} = (-500\cos 30°,\ 500\sin 30°) = (-433.0,\ 250.0)$ N. With CCW positive,

$$M_O = xF_y - yF_x = 0.4(250.0) - 0.3(-433.0) = 100.0 + 129.9 = 229.9\ \text{N}\cdot\text{m}\ \text{(CCW)}$$

Both terms turn the same way: the vertical component pushes up to the right of O, and the horizontal component pushes left above O.

(b) $d = M_O/F = 229.9/500 = 0.4598$ m.

(c) For a fixed point of application the smallest force is perpendicular to OA, with the whole distance $|OA| = \sqrt{0.4^2 + 0.3^2} = 0.5$ m as the arm: $F_{min} = 400/0.5 = 800.0$ N. OA points at $\tan^{-1}(0.3/0.4) = 36.87°$; for a clockwise moment the force must point 90° clockwise of that, at $36.87° - 90° = -53.13°$, that is 53.13° below $+x$. Check: $\vec{F} = (480, -640)$ N gives $M_O = 0.4(-640) - 0.3(480) = -400$ N·m. Good.



Solution 6

(a)

$$\vec{M}_O = \vec{r}_{OA} \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 0.4 & 0.6 & 0.2 \\ 60 & -40 & 120 \end{vmatrix}$$

$\hat{\imath}$: $0.6(120) - 0.2(-40) = 80.00$; $\hat{\jmath}$: $-[0.4(120) - 0.2(60)] = -36.00$; $\hat{k}$: $0.4(-40) - 0.6(60) = -52.00$.

$$\vec{M}_O = \{80.00\hat{\imath} - 36.00\hat{\jmath} - 52.00\hat{k}\}\ \text{N}\cdot\text{m}, \qquad M_O = 102.0\ \text{N}\cdot\text{m}$$

(b) $\hat{u}_a = (3, 4, 0)/5 = (0.6, 0.8, 0)$. Since O is on the axis, $M_a = \hat{u}_a \cdot \vec{M}_O$:

$$M_a = 0.6(80.00) + 0.8(-36.00) + 0 = 19.20\ \text{N}\cdot\text{m}, \qquad \vec{M}_a = 19.20\,\hat{u}_a = \{11.52\hat{\imath} + 15.36\hat{\jmath}\}\ \text{N}\cdot\text{m}$$

(c) $M_y = \hat{\jmath} \cdot \vec{M}_O = -36.00$ N·m, meaning 36.00 N·m about $-y$.

Solution 7

Position vector from a point on $-\vec{F}$ to a point on $\vec{F}$: $\vec{r}_{BA} = (0.2, 0.4, -0.3)$ m.

$$\vec{M}_1 = \vec{r}_{BA} \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 0.2 & 0.4 & -0.3 \\ 40 & -30 & 60 \end{vmatrix} = \{15.00\hat{\imath} - 24.00\hat{\jmath} - 22.00\hat{k}\} \text{ N} \cdot \text{m}$$

($\hat{\imath}$: $24 - 9$; $\hat{\jmath}$: $-(12 + 12)$; $\hat{k}$: $-6 - 16$.) Add $\vec{M}_2 = \{-25\hat{k}\}$ N·m, which is a free vector:

$$\vec{M}_R = \{15.00\hat{\imath} - 24.00\hat{\jmath} - 47.00\hat{k}\} \text{ N} \cdot \text{m}, \qquad M_R = 54.86 \text{ N} \cdot \text{m}$$

$\alpha = \cos^{-1}(15/54.86) = 74.13°$, $\beta = \cos^{-1}(-24/54.86) = 115.9°$, $\gamma = \cos^{-1}(-47/54.86) = 148.9°$.

Solution 8

x to the right, y up, CCW positive. The inclined load: $5\cos 60° = 2.500$ kN to the right and $5\sin 60° = 4.330$ kN down.

(a)

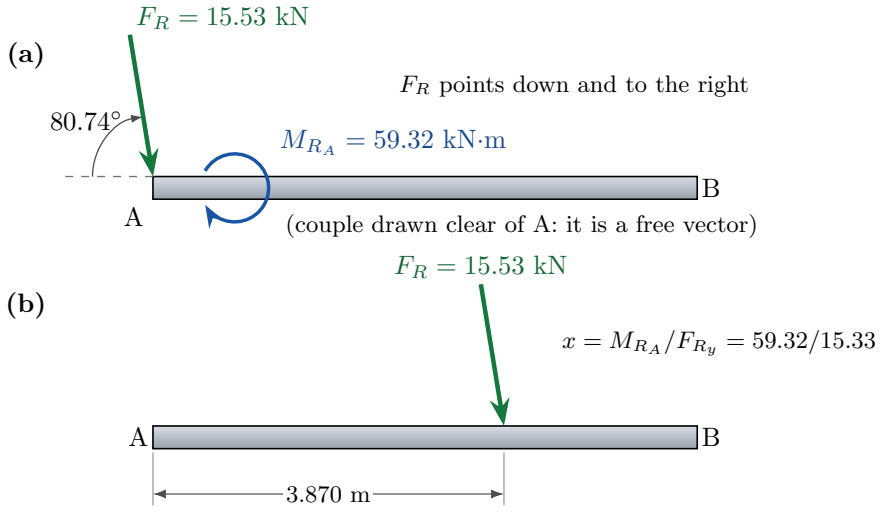
$$F_{Rx} = 2.500 \text{ kN}, \qquad F_{Ry} = -8 - 4.330 - 3 = -15.33 \text{ kN}$$
$$F_R = \sqrt{2.500^2 + 15.33^2} = 15.53 \text{ kN}, \quad \theta = \tan^{-1}(15.33/2.500) = 80.74° \text{ below } +x$$
$$M_{R_A} = -8(1.5) - 4.330(4) - 12 - 3(6) = -59.32 \text{ kN} \cdot \text{m}$$

So 59.32 kN·m clockwise. The 2.5 kN horizontal part acts along the beam axis through A and has no moment about A. The couple enters directly with no distance.

(b) Put $\vec{F}_R$ on the beam at distance x from A. Its moment about A is xF_{Ry}:

$$x(-15.33) = -59.32 \;\Rightarrow\; x = 3.870 \text{ m from A}$$

Sense check: a downward force to the right of A turns clockwise, which matches.



Solution 9

$F_R = -400 - 600 - 300 + 250 = -1050$ N, so 1050 N down. With all forces parallel to z the single resultant sits at the force-weighted average position (signs included):

$$\bar{x} = \frac{4(-400) + 4(-600) + 0(-300) + 2(250)}{-1050} = \frac{-3500}{-1050} = 3.333 \text{ m}$$

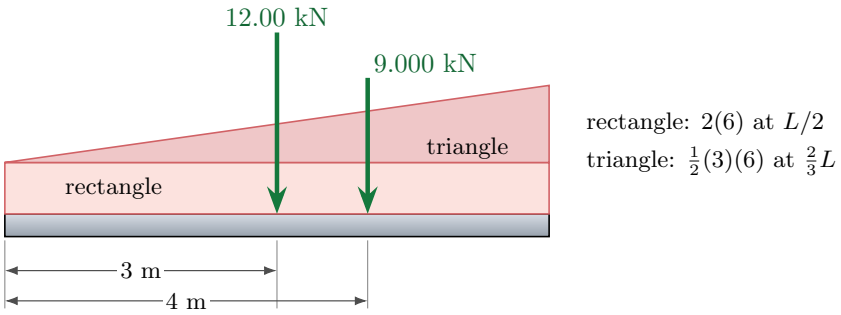
$$\bar{y} = \frac{0(-400) + 3(-600) + 3(-300) + 1.5(250)}{-1050} = \frac{-2325}{-1050} = 2.214 \text{ m}$$

Same thing via moments: $M_{R_x} = \sum yF = -2325$, $M_{R_y} = -\sum xF = 3500$ N·m, then $x = -M_{R_y}/F_R$, $y = M_{R_x}/F_R$. The downward loads alone would put the resultant at $(3.077, 2.077)$ m; the upward 250 N at $(2, 1.5)$ pushes it further away from that point.

Solution 10

(a) Split the trapezoid into a rectangle and a triangle. Rectangle: $2(6) = 12.00$ kN at 3 m. Triangle: $\frac{1}{2}(5-2)(6) = 9.000$ kN at $\frac{2}{3}(6) = 4$ m from A.

$$F_R = 21.00 \text{ kN}, \qquad \bar{x} = \frac{12(3) + 9(4)}{21} = 3.429 \text{ m from A}$$



(b)

$$F_R = \int_0^3 \tfrac{4}{9}x^2\,dx = \tfrac{4}{9}\cdot\tfrac{27}{3} = 4.000 \text{ kN}, \qquad \int_0^3 x\cdot\tfrac{4}{9}x^2\,dx = \tfrac{4}{9}\cdot\tfrac{81}{4} = 9.000 \text{ kN}\cdot\text{m}$$

$\bar{x} = 9.000/4.000 = 2.250$ m. Spandrel check: $n = 2$, $w_0 = 4$ kN/m, $L = 3$ m gives $F = w_0 L/3 = 4.000$ kN at $\frac{3}{4}L = 2.250$ m.

Solution 11

$$\vec{F}_R = \{300\hat{\imath} - 200\hat{\jmath} + 400\hat{k}\} \text{ N}, \qquad F_R = 538.5 \text{ N}, \qquad \hat{u}_{F_R} = (0.5571, -0.3714, 0.7428)$$

Moments about O: $(3,0,0)\times(0,0,400) = (0,-1200,0)$; $(0,2,0)\times(300,0,0) = (0,0,-600)$; $(3,2,0)\times(0,-200,0) = (0,0,-600)$.

$$\vec{M}_{R_O} = \{-1200\hat{\jmath} - 1200\hat{k}\} \text{ N}\cdot\text{m}$$

Component along $\vec{F}_R$: $M_{\parallel} = \vec{M}_{R_O}\cdot\hat{u}_{F_R} = (-1200)(-0.3714) + (-1200)(0.7428) = -445.7$ N·m. It is negative, so the wrench couple points opposite to $\vec{F}_R$ (a negative wrench), magnitude 445.7 N·m.

Perpendicular part: $\vec{M}_{\perp} = \vec{M}_{R_O} - M_{\parallel}\hat{u}_{F_R} = \{248.3\hat{\imath} - 1365.5\hat{\jmath} - 869.0\hat{k}\}$ N·m. Find P $= (x, y, 0)$ with $\vec{r}_{OP}\times\vec{F}_R = \vec{M}_{\perp}$, where $\vec{r}_{OP}\times\vec{F}_R = (400y,\ -400x,\ -200x - 300y)$:

$$400y = 248.3 \Rightarrow y = 0.6207 \text{ m}, \qquad -400x = -1365.5 \Rightarrow x = 3.414 \text{ m}$$

Check the $\hat{k}$ row: $-200(3.414) - 300(0.6207) = -869.0$. It matches, which confirms the arithmetic.